

CENTRAL BOARD OF SECONDARY EDUCATION
HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)
SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 6: HOW FORCES AFFECT MOTION

TIME ALLOWED: 45 MINUTES**MAXIMUM MARKS: 25****PAPER CODE: 086/CH06****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Which of the following is an effect that a balanced force system CAN produce on an object?**

- (A) Change in the speed of the object (B) Change in the direction of moving object
(C) Change in the shape and size of the object (D) Causing acceleration from rest

Q.2 The inertia of an object depends fundamentally and directly on its:

- (A) Velocity (B) Mass
(C) Volume (D) Acceleration

Q.3 When a moving passenger bus suddenly takes a sharp turn to the right, the passengers lean towards the left due to:

- (A) Inertia of rest (B) Inertia of motion
(C) Inertia of direction (D) Gravitational pull

Q.4 What is the momentum of an object of mass m moving with a velocity v ?

- (A) $(mv)^2$ (B) $m v^2$
(C) $\frac{1}{2} m v^2$ (D) $m v$

Q.5 What constant force is required to accelerate a 5 kg block from 4 m/s to 10 m/s in 3 seconds?

- (A) 10 N (B) 15 N
(C) 20 N (D) 30 N

Q.6 A cricket fielder pulls his hands backwards while catching a fast-moving ball. This action protects the fielder by:

- (A) Decreasing the time of impact to maximize force (B) Increasing the time of impact, thereby reducing the rate of change of momentum and impact force
(C) Increasing the momentum of the ball (D) Decreasing the mass of the ball

Q.7 According to Newton's Third Law of Motion, action and reaction forces:

- | | |
|---|---|
| (A) Act on the same body in the same direction | (B) Act on two different bodies in opposite directions simultaneously |
| (C) Act on the same body in opposite directions | (D) Cancel each other out completely so no motion occurs |

Q.8 A gun of mass 4 kg fires a bullet of mass 0.02 kg with a muzzle velocity of 400 m/s. What is the recoil velocity of the gun?

- | | |
|------------|------------|
| (A) -2 m/s | (B) -4 m/s |
| (C) -1 m/s | (D) -8 m/s |

Q.9 A 1000 kg car travelling at 20 m/s collides with a wall and comes to rest in 0.1 s. The average force exerted on the car during impact is:

- | | |
|--------------|-----------------|
| (A) 20,000 N | (B) 200,000 N |
| (C) 2,000 N | (D) 2,000,000 N |

Q.10 Why does a swimmer push water backwards with hands and feet to propel himself forward in a pool?

- | | |
|--------------------------------------|---|
| (A) Because water is denser than air | (B) According to Newton's third law, the water exerts an equal and opposite forward reaction force on the swimmer |
| (C) To increase his body weight | (D) To reduce water friction |

Q.11 Which famous Italian scientist first deduced that an object moving on a frictionless horizontal plane would continue to move with constant velocity indefinitely?

- | | |
|------------------|---------------------|
| (A) Isaac Newton | (B) Galileo Galilei |
| (C) Archimedes | (D) Johannes Kepler |

Q.12 If the net external unbalanced force acting on an object is zero, the acceleration of the object is:

- | | |
|-----------------------------|------------------------|
| (A) Continuously increasing | (B) Zero |
| (C) 9.8 m/s^2 | (D) Dependent on shape |

Q.13 When a carpet is beaten with a stick, dust particles fall out. This phenomenon is explained by:

- | | |
|---------------------------------------|-------------------------------------|
| (A) Inertia of rest of dust particles | (B) Inertia of motion of the carpet |
| (C) Newton's third law only | (D) Electrostatic attraction |

Q.14 A force of 5 N acts on a mass m_1 giving it an acceleration of 10 m/s^2 , and on mass m_2 giving it an acceleration of 20 m/s^2 . What acceleration would it give if both masses are tied together?

- | | |
|--------------------------|------------------------|
| (A) 6.67 m/s^2 | (B) 15 m/s^2 |
| (C) 5.0 m/s^2 | (D) 30 m/s^2 |

Q.15 The principle of rocket propulsion is primarily based on:

- | | |
|-----------------------------|--|
| (A) Conservation of energy | (B) Newton's third law and conservation of linear momentum |
| (C) Newton's first law only | (D) Pascal's principle |

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): A heavy iron ball has greater inertia than a football of identical size.

Reason (R): The inertia of an object is directly proportional to its mass.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Action and reaction forces never cancel each other out.

Reason (R): Action and reaction act on two completely different bodies.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Newton's second law gives a quantitative measurement of force.

Reason (R): Force is directly proportional to the rate of change of momentum.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Passengers in a car must wear safety seatbelts.

Reason (R): In a sudden collision, seatbelts provide an external force to restrain the passengers' bodies which tend to continue forward due to inertia of motion.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): When a bullet is fired from a rifle, the rifle recoils with the same acceleration as the bullet.

Reason (R): The force exerted on the rifle is equal and opposite to the force on the bullet.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: High Jump Safety and Collision Mechanics

In a school athletics track and field championship, athletes competing in the high jump event were instructed to land on a 50 cm thick foam cushion mattress. In previous decades, high jumpers landed on packed sand or hard ground. Physical training instructors demonstrated that landing on the thick foam cushion prevents severe spinal and limb injuries because the foam compresses gradually upon impact.

Q.21 What physical quantity remains the same when an athlete lands on hard ground versus on a foam mattress from a given height?

- (A) Impact force
 (B) Duration of collision
 (C) Total change in momentum of the athlete
 (D) Maximum compression distance

Q.22 How does the soft foam mattress reduce the impact force felt by the athlete?

- (A) By reducing the athlete's mass
 (B) By increasing the time duration taken for the athlete's momentum to become zero
 (C) By increasing gravity
 (D) By completely eliminating energy transformation

Q.23 If an athlete of mass 60 kg hits the cushion with a downward velocity of 6 m/s and comes to rest in 0.3 s, the average retarding force exerted by the cushion is:

- (A) 120 N (B) 1200 N
(C) 3600 N (D) 600 N

Q.24 If the same athlete had landed on hard concrete stopping in just 0.01 s, the impact force would have been:

- (A) 1,200 N (B) 12,000 N
(C) 36,000 N (D) 360 N

Q.25 Which similar practical safety mechanism utilizes this same impulse-momentum principle?

- (A) Automobile air bags (B) Packing fragile glass items in bubble wrap
(C) Shock absorbers in motorcycles (D) All of the above

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(C)	Balanced forces have zero net force, so they cannot accelerate an object, but equal opposite forces can compress or distort its shape (e.g., squeezing a rubber ball).
Q.2	(B)	Mass is the quantitative measure of inertia. A heavier body has greater resistance to changes in its state of motion.
Q.3	(C)	Due to inertia of direction, passenger bodies tend to maintain their original straight-line path when the bus turns right.
Q.4	(D)	Linear momentum is defined as the product of mass and velocity ($p = mv$), with SI unit $\text{kg} \cdot \text{m/s}$.
Q.5	(A)	$a = \frac{10 - 4}{3} = 2 \text{ m/s}^2$. $F = ma = 5 \times 2 = 10 \text{ N}$.
Q.6	(B)	From $F = \frac{\Delta p}{\Delta t}$, increasing duration Δt drastically reduces the impact force F felt by the palms.
Q.7	(B)	Action and reaction forces always act on two different interacting objects simultaneously with equal magnitude and opposite direction.
Q.8	(A)	$m_1 v_1 + m_2 v_2 = 0 \Rightarrow 4 \times v_{\text{gun}} + 0.02 \times 400 = 0 \Rightarrow 4 v_{\text{gun}} = -8 \Rightarrow v_{\text{gun}} = -2 \text{ m/s}$.
Q.9	(B)	$F = \frac{m(v - u)}{t} = \frac{1000(0 - 20)}{0.1} = -200,000 \text{ N}$.
Q.10	(B)	Pushing water backwards (action) produces an equal and opposite forward push (reaction) from the water on the swimmer.
Q.11	(B)	Galileo studied motion on double inclined planes and concluded that an object in motion continues moving unless resisted by friction.
Q.12	(B)	By $F = ma$, if $F_{\text{net}} = 0$, then $a = 0$. The object maintains constant velocity or remains at rest.
Q.13	(A)	The carpet moves forward when struck, but dust particles tend to remain in their state of rest due to inertia and fall away.
Q.14	(A)	$m_1 = 5/10 = 0.5 \text{ kg}$, $m_2 = 5/20 = 0.25 \text{ kg}$. Combined mass $M = 0.75 \text{ kg}$. $a = F/M = 5/0.75 = 6.67 \text{ m/s}^2$.
Q.15	(B)	Expulsion of exhaust gases at high speed downward imparts an equal upward thrust (momentum conservation / action-reaction).
Q.16	(A)	Mass determines inertia. Because iron is much denser and more massive than a football, it offers greater resistance to acceleration.
Q.17	(A)	Forces cancel only when acting on the same single body. Action and reaction act on different interacting bodies.
Q.18	(A)	The formula $F = ma$ derived from Newton's second law provides the mathematical definition and metric value of force.
Q.19	(A)	When a car stops abruptly, inertia carries unbelted occupants forward into windshield or steering wheel; seatbelts supply the decelerating force.
Q.20	(D)	Reason is true (forces are equal and opposite). But Assertion is false because $a = F/m$; since the rifle is much more massive, its acceleration is far smaller than the bullet's.
Q.21	(C)	The athlete has the same mass and reaches the ground with the same impact velocity, so the change in momentum $\Delta p = 0 - mv$ is identical in both cases.
Q.22	(B)	Foam compresses over time Δt . Because $F = \frac{\Delta p}{\Delta t}$, increasing impact time substantially lowers impact force.
Q.23	(B)	$F = \frac{m \Delta v}{\Delta t} = \frac{60 \times 6}{0.3} = 1200 \text{ N}$.

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.24	(C)	$F = \frac{60 \times 6}{0.01} = 36,000 \text{ N}$ (30 times higher, enough to fracture bones!).
Q.25	(D)	Airbags, bubble wrap, and shock absorbers all increase collision contact duration to minimize peak impact force.